Half-line completion of the canonical Bell slack

A quadratic selector on each Bob half-line certifies a rational canonical source that no global polynomial selector can certify. The additional freedom has a simple Gram representation: a quadratic matrix positive on $v \geq 0$ is a linear matrix square plus v times a constant positive matrix. Applied to the canonical completion, this gives four constant positive semidefinite blocks and a closed source cone. We give the complete affine pencil and a rational certificate exhibiting the gain.

1 The canonical completion

Let $c = (d, \lambda, n, m, C, A, D, \alpha, \beta) \geq 0$ and $D_v = 1 + v^2$. The canonical state change collects the source into four quadratic matrices:

$$P(v) = \begin{pmatrix} 4mv^2 & -4mv \\ -4mv & 4m + (4\beta + 2d)D_v \end{pmatrix}, \qquad N(v) = \begin{pmatrix} 4CD_v + 4D + 2d & (4D + 2d)v \\ (4D + 2d)v & (4D + 2d)v^2 \end{pmatrix},$$
$$E(v) = \begin{pmatrix} 4AD_v + (4\alpha + 2d)v^2 & -(4\alpha + 2d)v \\ -(4\alpha + 2d)v & 4\alpha - 2dv^2 \end{pmatrix}, \qquad K(v) = 4\lambda \begin{pmatrix} 1 & v \\ v & v^2 \end{pmatrix} + 4nD_v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

For $U = u^2$ and independent $a, b \in \mathbb{R}^2$, the reduced state form is

$$\Phi_c = (1 + U)a^T P a + U(1 + U)b^T N b + U(a - b)^T E(a - b) + (a + Ub)^T K(a + Ub). \tag{1}$$

The substitution

$$W = a_1 + Ub_1, \quad X = a_2 + Ub_2, \quad Y = u(a_1 - b_1), \quad Z = u(a_2 - b_2)$$

gives the complete canonical parent $F_c = (1 + u^2)\Phi_c$. It is invertible when $u \neq 0$, with determinant $u^2(1 + u^2)^2$. Thus positivity of this form covers all state directions; the finite axis follows by polynomial continuity. Homogenizing the parent to bidegree $(2, 2)$ in the two projective angle pairs gives the infinite endpoints. Real state positivity also covers complex states by separating their real and imaginary parts.

A real 2×2 selector J gives the two completion matrices

$$G_J = \begin{pmatrix} P + K & J \\ J^T & N + K \end{pmatrix}, \qquad H_J = \begin{pmatrix} P + E & K - E - J \\ K - E - J^T & N + E \end{pmatrix}. \tag{2}$$

Their role is the exact cancellation

$$\Phi_c = \begin{pmatrix} a \\ Ub \end{pmatrix}^T G_J \begin{pmatrix} a \\ Ub \end{pmatrix} + U \begin{pmatrix} a \\ b \end{pmatrix}^T H_J \begin{pmatrix} a \\ b \end{pmatrix}. \tag{3}$$

We seek $J_+(v) = J_0 + vJ_1 + v^2 J_2$ making both matrices positive semidefinite for $v \geq 0$. Their parity makes this sufficient for all Bob angles: with $T = \mathrm{diag}(1, -1)$, each of P, N, E, K satisfies $M(-v) = TM(v)T$. For $v < 0$ set $J_-(v) = TJ_+(-v)T$. Congruence by $\mathrm{diag}(T, T)$ then restores both inequalities on that half-line. At zero use $J_+(0)$; no continuity of the selector is needed.

2 Four constant Gram blocks

Lemma 1. *A real symmetric 4×4 quadratic $M(v)=M_0+vM_1+v^2M_2$ is positive semidefinite on $v\ge0$ if and only if some unrestricted 4×4 matrix B satisfies*

$$S=\begin{pmatrix}M_0 & B\\ B^T & M_2\end{pmatrix}\succeq 0, \qquad T_M=M_1-B-B^T\succeq 0.$$

Proof. Sufficiency follows from $M(v)=[I\ vI]S[I\ vI]^T+vT_M$. For necessity, $M(t^2)$ is positive on the real line and has degree at most four. Matrix Fejér–Riesz gives polynomial factors of degree at most two, also for singular matrices. Taking real and imaginary parts and averaging under $t\mapsto -t$ gives

$$M(v)=(R_0+vR_2)^T(R_0+vR_2)+vR_1^TR_1.$$

Take $S=[R_0\ R_2]^T[R_0\ R_2]$ and $T_M=R_1^TR_1$. The degree bounds are sharp: vI requires the half-line term and v^2I requires a linear factor. No coefficient is divided out. The assertion transfers to real closed fields because its finite Gram formulation is first order. The real-line factorization used here is [1, Theorem 1.2]. □

Write G_j, H_j for the coefficients of v^j in (2), with $J=J_0+vJ_1+v^2J_2$. They are affine in c and the selector entries. Introducing two unrestricted 4×4 matrices B_G, B_H gives the pencil

$$\boxed{\begin{aligned} &S_G=\begin{pmatrix}G_0 & B_G\\ B_G^T & G_2\end{pmatrix}\succeq 0, \qquad S_H=\begin{pmatrix}H_0 & B_H\\ B_H^T & H_2\end{pmatrix}\succeq 0,\\ &T_G=G_1-B_G-B_G^T\succeq 0, \quad T_H=H_1-B_H-B_H^T\succeq 0.\end{aligned}} \tag{4}$$

Lemma 1 proves both inclusions between this projection and the quadratic half-line selector cone. It also retains every singular case and every zero source face. There are $12+16+16=44$ auxiliary scalars and no remaining coefficient equations. The block orders $8,8,4,4$ total 24; adding the nine scalar blocks $[c_i]$ gives total order 33.

There is an explicit strictly positive dual certificate. Set

$$W_S=\begin{pmatrix}I_4 & I_4/2\\ I_4/2 & I_4\end{pmatrix}, \qquad W_T=I_4/2.$$

Every auxiliary coefficient cancels in the exact identity

$$\langle W_S, S_G+S_H\rangle+\langle W_T, T_G+T_H\rangle=16\sum_i c_i, \tag{5}$$

where $\langle A,B\rangle=\operatorname{tr}(A^TB)$. Indeed the B traces cancel, the middle coefficients G_1, H_1 have zero diagonal, and $\operatorname{tr}S_G+\operatorname{tr}S_H=16\sum c_i$. Both weights have minimum eigenvalue $1/2$, so feasibility implies

$$\operatorname{tr}S_G+\operatorname{tr}S_H+\operatorname{tr}T_G+\operatorname{tr}T_H\le 32\sum_i c_i.$$

All B_G, B_H, J_0, J_2 entries are off-diagonal Gram entries. For a J_1 entry at distinct G_1 indices r,t, combining the local PSD entry bounds in $G_1=T_G+B_G+B_G^T$ gives $|(J_1)_{ij}|\le(\operatorname{tr}T_G+\operatorname{tr}S_G)/2$. Consequently every feasible auxiliary vector satisfies

$$\|y\|_\infty\le 16\sum_i c_i. \tag{6}$$

Bounded subsequences now prove the projection closed; replacing $\sum c_i$ by $\sum|c_i|$ covers unrestricted source inputs. The positive weights in (5) give strict feasibility of the dual auxiliary equations. Fixed-source feasibility cannot be weakly infeasible: an unbounded near-feasible sequence, normalized by its auxiliary norm, would give feasible source-zero data with a nonzero auxiliary vector, contradicting (6); a bounded sequence gives a feasible limit.

3 An exact source rescued by the half-line completion

Take

$$c_* = \left(359, 0, 0, \frac{75}{16}, 60, 416, \frac{41}{2}, \frac{831}{2}, \frac{151}{2}\right).$$

Dividing the source and every matrix below by 359 gives $d = 1$. An exact selector on $v \geq 0$ is

$$J_+(v) = \frac{1}{64}\begin{pmatrix} 3043v - 2109v^2 & 3875v^2 \\ 1403 + 23874v + 2306v^2 & 815v + 43455v^2 \end{pmatrix}. \tag{7}$$

The two remaining auxiliary matrices are

$$B_G = \frac{1}{256}\begin{pmatrix} 0 & 0 & 0 & 0 \\ -4800 & -48015 & -27346 & 3260 \\ 12172 & 186862 & -21340 & 204800 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{8}$$

$$B_H = \frac{1}{18496}\begin{pmatrix} -6910952 & 240550 & 5277344 & 1292578 \\ -41117866 & -4908665 & 35528504 & 2316913 \\ 3043153 & 1489795 & -7316324 & 276862 \\ 36194938 & -415582 & -30578223 & -3325523 \end{pmatrix}. \tag{9}$$

Together with (2) and (4), these rational matrices specify the certificate completely. Their positive semidefiniteness is verified by the exact elimination in the appendix. Thus (3) proves the full source valid for all states and all testing angles.

The difference from a global polynomial completion is essential. If a polynomial $J(v)$ satisfies (2) on the real line, its entries have degree at most two: each squared entry is bounded by a product of two quadratic diagonal entries of G_J. Parity averaging makes its diagonal entries even and its off-diagonal entries odd. On the face $\lambda = n = 0$, the zero diagonals of $G_J(0)$ and $|J_{12}(v)|^2 \leq 4m(4D + 2d)v^4$ then reduce it to

$$J(v) = \begin{pmatrix} pv^2 & 0 \\ rv & sv^2 \end{pmatrix}.$$

For this source the exact contact is $U = 1/4$, $v = 2$, $a = (17, 2)^T$, $b = (12, -18)^T$. The nonnegative terms in (3) vanish there, forcing $J(2)b = -4P(2)a$ and $J(2)^T a = -N(2)b/4$. These equations uniquely give $(p, r, s) = (-100, 2000, 1200)$. At $v = 1$ their G_J principal submatrix on (a_2, b_1, b_2) is

$$\begin{pmatrix} 8235/4 & 2000 & 1200 \\ 2000 & 1280 & 800 \\ 1200 & 800 & 800 \end{pmatrix}, \qquad \det = -412640000 < 0.$$

Consequently no global polynomial selector works for this source. The reflected selector (7) instead contains $|v|$ and $v|v|$: its upper entries are $(3043|v| - 2109v^2)/64$ and $3875v|v|/64$; for $v \neq 0$ the lower-left entry also contains $1403\,\mathrm{sign}(v)/64$.

The result is an exact closed affine cone of quadratic half-line selectors, strictly more effective than global polynomial selectors on this example. It is a proper subcone of the valid canonical sources: the accompanying separating certificate excludes an all-positive rational source. Allowing the additional selector terms $\sqrt{v}$ and $v^{3/2}$ certifies that source.

A Exact positivity of the rational Gram matrices

For the source above, $K = 0$ and the three nonzero source matrices are

$$P = \begin{pmatrix} 75v^2/4 & -75v/4 \\ -75v/4 & 4155/4 + 1020v^2 \end{pmatrix},$$

$$N = \begin{pmatrix} 1040 + 240v^2 & 800v \\ 800v & 800v^2 \end{pmatrix}, \qquad E = \begin{pmatrix} 1664 + 4044v^2 & -2380v \\ -2380v & 1662 - 718v^2 \end{pmatrix}.$$

Use (4) with (7)–(9). Unpivoted symmetric elimination gives $M = LDL^T$ with unit diagonal in L. At a positive residual pivot d_i, set $L_{ji} = R_{ji}/d_i$ and update $R_{jk} \leftarrow R_{jk} - R_{ji}R_{ik}/d_i$ for $j, k > i$. At every zero pivot the remaining row vanishes exactly, so it is left unchanged. The following table lists all zero positions (numbered from one) and the floors of the positive rational pivots, in order. Every positive pivot lies in $[k, k+1)$ for its printed integer k.

M	zero positions	floors of positive pivots	rank M
S_G	$1, 4, 8$	1038, 1039, 16, 382, 162	5
S_H	8	1664, 2700, 1039, 639, 1562, 214, 67	7
T_G	$1, 3, 4$	375	1
T_H	4	747, 483, 430	3

For example, $T_G = (5335/128)(0, 3, -2, 0)^T(0, 3, -2, 0)$. The printed rational inputs and this recurrence give a pen-and-paper positivity test including all singular directions. The accompanying exact witness file also supplies every entry of L, D; the rational checker recomputes them from these inputs and verifies $M = LDL^T$.

References

[1] A. Zalar, *Matrix Fejér–Riesz theorem with gaps*, Theorem 1.2, arXiv:1503.06034.